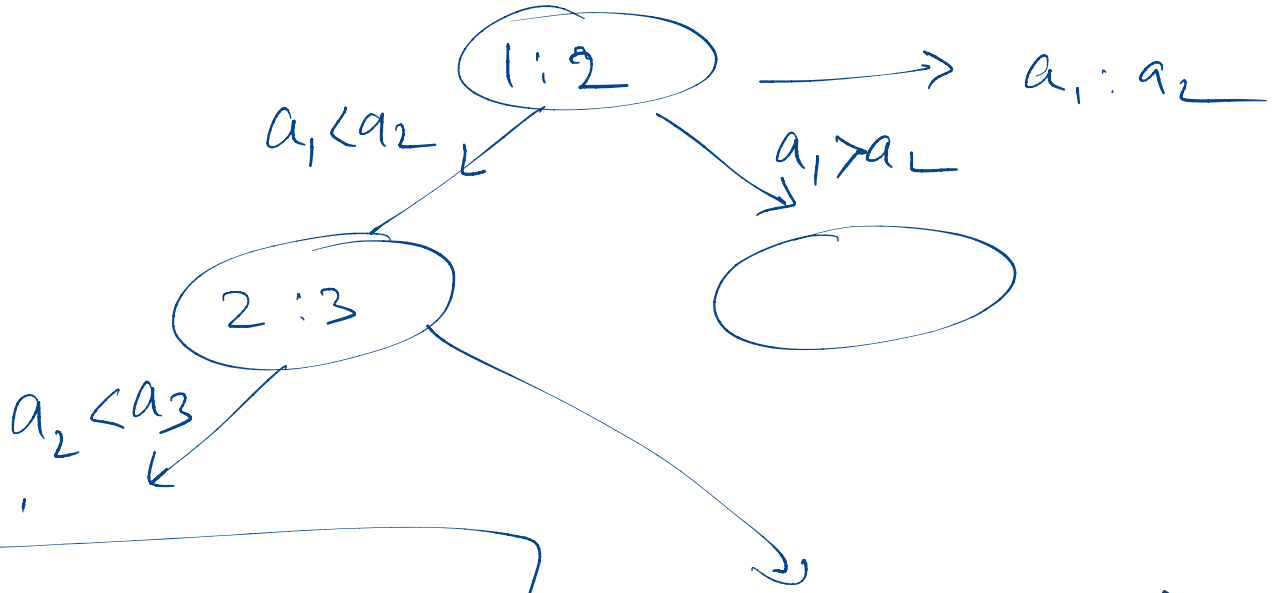
Decision Tree

a_1, a_2, a_3



$a_1 < a_2$ & $a_2 < a_3$

$\Rightarrow a_1 < a_2 < a_3$

$\Rightarrow a_1 < a_3$

$a_1 < a_2$ $\quad a_2 > a_3$

$\not\Rightarrow a_1 > a_3$

Not true

Prefix sum

$A = \langle a_1, a_2, \dots a_n \rangle$

$S = \langle s_1, s_2, \dots s_n \rangle$

$$s_i = \sum_{k=1}^{i} a_k = s_{i-1} + a_i$$

$A = [3, 7, 8, 0, 1]$

$S = [3, 10, 18, 18, 19] \rightarrow$ Prefix sum